Santa Cruz Community Solar Project Documents

PDF Document 1: Project Narrative and Technical Specifications

COASTAL COMMUNITY SOLAR INITIATIVE

Harvey West Park Solar Array

Prepared for: County of Santa Cruz
Prepared by: Pacific Coast Renewables, Inc.
Date: October 1, 2025

Executive Summary

Pacific Coast Renewables, Inc. is pleased to present this comprehensive project narrative for the proposed Harvey West Park Community Solar Array. This initiative represents a significant step forward in Santa Cruz County's commitment to renewable energy and community resilience. The project will be sited on approximately 2.8 acres of underutilized land adjacent to the Harvey West Park baseball fields, transforming this space into a productive solar energy facility while maintaining the park's recreational character.

The solar array is designed to serve approximately 95 local households and small businesses through a community solar subscription model. This approach democratizes access to solar energy, particularly benefiting renters, condominium owners, and those with shaded or unsuitable roofs. By participating in this program, subscribers will receive credits on their Pacific Gas & Electric (PG&E) bills corresponding to their allocated share of the system's production, typically resulting in 10-15% savings on electricity costs.

Our partnership with the County of Santa Cruz reflects a shared vision for sustainable energy infrastructure. Under the terms of our ground lease agreement, Pacific Coast Renewables will lease the designated parcel for an initial term of twenty-five years, with lease payments structured to provide incremental revenue growth to the County's Parks Department. Initial annual payments will commence at $6,500, escalating by $750 every five years, providing predictable funding that can be reinvested in park improvements and maintenance.

Site Characteristics and Selection Rationale

Harvey West Park, located at 326 Coral Street in Santa Cruz, encompasses 23 acres of recreational facilities including ball fields, a skate park, and open space. The proposed solar array site sits on the western perimeter of the park, an area currently characterized by ruderal vegetation and minimal recreational use. Site selection was guided by several key factors: excellent solar access with minimal shading, proximity to existing electrical

infrastructure, sufficient distance from primary recreational areas to avoid conflicts, and soil conditions suitable

infrastructure, sufficient distance from primary recreational areas to avoid conflicts, and soil conditions suitable for driven-pile foundations.

Preliminary geotechnical investigations conducted by Central Coast Engineering Solutions revealed predominantly sandy loam soils with adequate bearing capacity for our proposed racking system. The site exhibits a gentle slope of approximately 3-5% from north to south, which will require minimal grading to establish proper drainage patterns. Environmental surveys completed in spring 2025 confirmed the absence of sensitive species or habitats within the project footprint, though we will implement standard erosion control measures during construction to protect the adjacent riparian corridor of Branciforte Creek, located approximately 200 feet to the west.

Technical Design Overview

The Harvey West Park array will utilize cutting-edge solar technology optimized for the Central Coast's marine climate. We have selected Canadian Solar HiKu7 modules, which feature advanced mono-PERC cell technology and superior performance in diffuse light conditions—a critical consideration given Santa Cruz's frequent morning fog. Each module produces 450 watts under standard test conditions, and we will deploy 1,556 total modules across the site to achieve a system capacity of 700 kilowatts DC.

Module orientation and tilt angle have been carefully optimized through extensive modeling using PVsyst software. Our design features a south-facing orientation at a 20-degree tilt angle, which balances optimal annual energy production with aesthetic considerations and wind load minimization. This relatively low tilt angle also provides clearance benefits, allowing us to maintain adequate spacing between rows while minimizing the system's visual profile as viewed from adjacent park areas.

The electrical architecture employs a distributed inverter topology, utilizing eight Fronius Symo Advanced 75 kW string inverters strategically positioned throughout the array. This distributed approach offers several advantages over traditional central inverter systems, including enhanced system reliability (a single inverter failure affects only a portion of the array), improved monitoring granularity, and reduced DC cable runs. Each inverter will be housed in a weather-resistant NEMA 3R enclosure mounted on a concrete pad measuring 4 feet by 3 feet.

Power collection from the inverters occurs through a 480-volt AC collection system, with all wiring routed underground in PVC conduit to maintain the site's aesthetic integrity and protect against damage. The collection system converges at a pad-mounted utility transformer rated at 500 kVA, which steps voltage up to 12 kilovolts for interconnection with PG&E's distribution system. The point of interconnection is located at an existing PG&E pole approximately 180 feet northeast of the array, requiring directional boring beneath the park access road to install the necessary 15kV feeder cable.

Structural and Mounting Systems

Our structural approach prioritizes both engineering integrity and environmental sensitivity. The GameChange

Solar Genius Tracker™ fixed-tilt system has been selected for its proven performance in coastal environments and certification to withstand exposure category C wind loads up to 120 miles per hour. The system employs hot-dip galvanized steel posts driven approximately 6-8 feet into the ground, eliminating the need for concrete foundations and minimizing site disturbance.

Each row of modules will be supported by a torque-tube rail system, with modules mounted in a landscape (horizontal) orientation, two panels high. This configuration provides optimal structural efficiency while maintaining a maximum height of 11 feet at the top of the tilted modules and a minimum clearance of 4 feet at the bottom edge. The elevated clearance serves multiple purposes: facilitating air circulation to enhance module cooling, allowing for vegetation growth beneath the array to prevent erosion, and providing access for maintenance personnel and equipment.

Row spacing has been calculated using industry-standard solar pathfinder analysis and modeling tools to prevent inter-row shading during peak sun hours. Our design maintains a pitch (center-to-center distance) of 16 feet between module rows, ensuring zero shading at solar noon even during the winter solstice. This spacing strikes an optimal balance between land use efficiency and energy production maximization.

Energy Production and Performance Expectations

Detailed production modeling conducted using NREL's System Advisor Model (SAM) and PVsyst indicates the Harvey West array will generate approximately 1,190,000 kilowatt-hours annually under typical meteorological conditions. This figure incorporates realistic loss factors including soiling (a particular concern in our fog-prone climate), module degradation, DC and AC system losses, inverter inefficiencies, and expected availability factors.

On a monthly basis, production will vary seasonally, with peak output occurring during the summer months of June through August when we anticipate generating 130,000-140,000 kWh per month. Winter production during December and January will be approximately 60-70,000 kWh monthly, reflecting shorter day lengths and increased cloud cover. These seasonal variations are well-understood by our subscriber management team and are clearly communicated to program participants during the enrollment process.

System monitoring will be comprehensive, utilizing both inverter-level production tracking and module-level monitoring through integrated power optimizers. Real-time data on system performance, environmental conditions, and equipment status will be accessible through a web-based portal, providing transparency to both Pacific Coast Renewables operations staff and County sustainability coordinators. This monitoring infrastructure also enables rapid fault detection and resolution, minimizing downtime and production losses.

Construction Methodology and Timeline

We anticipate a construction duration of approximately 18-20 weeks from notice to proceed through final commissioning. The construction sequence will begin with site preparation activities including surveying,

staking, erosion control installation, and access road improvements. Pile driving operations will follow, utilizing specialized equipment that can install foundation posts at a rate of 50-80 per day depending on soil conditions.

Module racking installation and array assembly will occur concurrently with electrical infrastructure work, including trenching for underground conduits, installation of inverter pads and equipment, and transformer pad preparation. All work will be conducted during daylight hours (7:00 AM to 5:00 PM, Monday through Friday) to minimize impacts on park users and adjacent residential areas. We have coordinated with County Parks staff to schedule construction during the fall and winter months when park utilization is typically lower.

The electrical interconnection work, which requires coordination with PG&E, represents a critical path item in our schedule. We have already initiated preliminary discussions with PG&E's Generation Interconnection group and submitted our application under the Wholesale Distribution Tariff (WDT) process. Based on PG&E's initial screening, the project has been assigned to the Fast Track processing queue, indicating that no distribution system upgrades are anticipated to be required for interconnection.

Environmental Stewardship and Site Management

Pacific Coast Renewables is committed to environmental stewardship throughout all phases of the project lifecycle. During construction, we will implement a comprehensive Stormwater Pollution Prevention Plan (SWPPP) developed in accordance with California's Construction General Permit requirements. This plan includes erosion control measures such as silt fencing, fiber rolls, and stabilized construction entrances to prevent sediment discharge to Branciforte Creek and municipal storm drain systems.

Post-construction site management emphasizes low-impact landscaping approaches that enhance both ecological function and visual aesthetics. The area beneath and between module rows will be vegetated with a carefully selected mix of native and climate-appropriate species including California fescue, purple needlegrass, and yarrow. This vegetation provides erosion control, habitat value for pollinators and small wildlife, and dust suppression while requiring minimal irrigation after establishment.

Perimeter security fencing will consist of 7-foot chain-link with vinyl privacy slats in a dark green color to blend with the park environment. Two access gates will be provided: a 16-foot wide main gate on the north side for equipment access and a 4-foot wide personnel gate on the east side. Signage will be installed per County requirements, including clearly visible warnings for electrical hazards and facility identification.

PDF Document 2: Contractor Information and Electrical Design Details

HARVEY WEST SOLAR ARRAY - CONTRACTOR PACKAGE

Project Developer: Pacific Coast Renewables, Inc.

Pacific Coast Renewables operates throughout California's Central Coast region, specializing in commercial-scale and community solar installations. Our Santa Cruz office, located at 2340 Mission Street, Suite 205, serves as the local hub for project development and ongoing operations. The company holds an active C-46 Solar Contractor license (License #1047392) and maintains all required bonding and insurance as prescribed by the Contractors State License Board.

Jennifer Park serves as our Director of Project Development and will act as the primary point of contact for this initiative. Jennifer brings 12 years of experience in renewable energy project management, with particular expertise in navigating California's complex regulatory environment. She can be reached at (831) 555-0847 or via email at j.park@pacificcoastrenewables.com. Our office maintains regular business hours Monday through Friday, 8:30 AM to 5:00 PM, with emergency contact protocols established for after-hours construction or operational issues.

The company's track record in Santa Cruz County includes successful completion of three previous ground-mount solar installations totaling 1.8 megawatts, as well as numerous rooftop commercial installations for local businesses and institutions. References from these projects, including contact information for facility managers and municipal officials, are available upon request.

Electrical Contracting and Engineering

Electrical design and construction services will be provided by Monterey Bay Electric Systems, a well-established firm with deep expertise in renewable energy installations. Monterey Bay Electric holds a California C-10 Electrical Contractor license (#E-876543) and has been a trusted partner on numerous Pacific Coast Renewables projects over the past eight years. Their principal office is located at 1556 Industrial Road in Watsonville, approximately 15 miles from the project site, enabling rapid response for both construction support and future service needs.

The project's electrical engineer of record is Marcus Chen, PE, a licensed professional electrical engineer in California (License #E-67234) with specialization in photovoltaic systems and utility interconnection. Marcus has prepared the detailed electrical calculations, equipment specifications, and protection coordination studies required for PG&E interconnection approval. His stamped and signed engineering drawings form part of the complete submittal package being provided to the County of Santa Cruz Building Department.

David Ramirez serves as Monterey Bay Electric's project superintendent and will oversee all field electrical work. David is a California Certified General Electrician and holds specialized certifications in solar PV installation from the North American Board of Certified Energy Practitioners (NABCEP). His crew includes four journey-level electricians, all of whom have completed at least three utility-scale solar projects and are current on OSHA 30-hour construction safety training. Contact information for David: (831) 555-0923 or d.ramirez@montereybayelectric.com.

Detailed Electrical System Architecture

The Harvey West array's electrical design reflects current best practices for utility-interactive solar installations while incorporating specific requirements of PG&E's interconnection standards and California's Rule 21. Understanding this system requires examining both the DC (direct current) generation side and the AC (alternating current) utility interface side.

On the DC side, the 1,556 solar modules are organized into 97 separate strings, each containing 16 modules connected in series. This series configuration produces a string voltage of approximately 640 volts DC under standard test conditions, well within the operating window of our selected Fronius inverters which can accommodate DC input voltages from 500 to 1,000 volts. String currents are limited by the modules' maximum power point current of 11.36 amperes, providing appropriate safety margins for all DC wiring and overcurrent protection devices.

These 97 strings are distributed across eight DC combiner boxes, strategically positioned throughout the array to minimize cable runs and voltage drop. Each combiner box consolidates 12-13 strings, employing appropriately rated string fuses (15 ampere, 1000 volt DC rated) for overcurrent protection. The combiners also incorporate DC-rated disconnects and surge protection devices to safeguard against lightning and switching transients. Combiner box locations have been carefully chosen to maintain cable runs under 200 feet to any given string, thereby keeping voltage drop below the 2% threshold specified in the National Electrical Code.

DC wiring throughout the array consists of USE-2 or PV Wire rated for 90°C and 2000 volts, providing substantial safety margins above the system's maximum voltage. String wiring uses 10 AWG conductors, while homerun cables from combiners to inverters employ larger 4 AWG or 6 AWG conductors depending on current and distance. All DC wiring will be installed in above-ground cable tray or direct-buried HDPE conduit, with burial depth of 24 inches per NEC requirements for renewable energy systems.

The eight Fronius Symo Advanced inverters represent the heart of the system's power conversion infrastructure. Each inverter accommodates input from approximately 12 strings (194 modules) and converts the DC power to 480-volt, three-phase AC power synchronized with PG&E's grid. These inverters feature transformer-less topology, achieving weighted efficiency ratings exceeding 98%, which translates to minimal power losses during the conversion process.

Inverter placement follows a distributed topology with units positioned at approximately even intervals throughout the array. This approach provides several operational advantages: reduced AC collection losses due to shorter cable runs to the main transformer, enhanced troubleshooting capabilities since each inverter serves a discrete array section, and improved reliability through redundancy—if one inverter requires service, 87.5% of the system remains operational. Each inverter location includes a readily accessible AC disconnect switch mounted adjacent to the inverter, facilitating safe maintenance and meeting first responder safety requirements.

The AC collection system gathers power from the eight inverters through underground three-phase circuits routed in 2-inch PVC conduit. These collection circuits employ 2 AWG aluminum conductors with THWN-2 insulation rated for 90°C operation in wet locations. Six of the inverters connect to a primary AC collection

circuit, while the remaining two connect to a secondary circuit, both terminating at a main AC disconnect/combiner enclosure located near the pad-mounted transformer.

This main AC combiner serves multiple functions: it provides a central disconnection point for the entire array, houses monitoring equipment and protection relays, and includes metering current transformers for production monitoring. The enclosure, rated NEMA 3R for outdoor installation, contains appropriately sized circuit breakers for each inverter feeder (100 ampere, 3-pole) and a main output breaker (500 ampere, 3-pole) feeding the transformer primary.

Utility Interconnection Infrastructure

The pad-mounted transformer represents the interface between the project's 480-volt AC collection system and PG&E's 12.47 kilovolt distribution network. This 500 kVA three-phase transformer incorporates several features specific to distributed generation applications. Primary-side fusing provides overcurrent protection per PG&E specifications, while the transformer itself includes no-load tap changers allowing ±2.5% voltage adjustment to optimize power delivery under varying grid conditions.

The transformer pad, constructed of reinforced concrete measuring 10 feet by 8 feet, includes secondary containment capable of holding 110% of the transformer's oil volume in the unlikely event of a leak. A 4-inch conduit stubbed into the pad accommodates the 480-volt AC input cables from the main combiner enclosure, while a separate 6-inch conduit provides for the 15kV output cable to the utility interconnection point.

Between the transformer and PG&E's distribution system, we will install a dedicated utility-accessible disconnecting means, specifically a load-break elbow termination at the utility pole. This configuration, standard for distributed generation interconnections in PG&E territory, allows utility personnel to isolate the solar facility without requiring access to the gated project site. The 15kV interconnection cable consists of single-conductor, shielded medium-voltage cable rated at 15kV (meeting PG&E's 133% voltage rating requirement for 12.47kV systems), installed in a 4-inch HDPE conduit via directional boring beneath the park access road.

Protection and control systems meet PG&E's requirements as specified in Electric Rule 21, California's interconnection standard. Primary protection includes an SEL-351 microprocessor-based protection relay configured to detect and respond to fault conditions including overcurrent, overvoltage, undervoltage, over/under frequency, and unintentional islanding. The relay connects to the main AC combiner breaker as the primary disconnection device and communicates status information to both the site SCADA system and, if required by PG&E, to the utility's distribution management system via cellular modem.

The inverters provide secondary protection functions including anti-islanding detection per IEEE 1547 and UL 1741-SA standards, voltage and frequency ride-through capabilities, and reactive power control. These "smart inverter" functions enable the array to support grid stability by providing voltage regulation through var injection/absorption and frequency response through active power curtailment when required. Pacific Coast

injection/absorption and frequency response through active power curtailment when required. Pacific Coast

Renewables has programmed these functions in accordance with California's Rule 21 Smart Inverter Implementation Guidelines, with settings verified during commissioning and documented for PG&E records.

System Monitoring and Control

Comprehensive monitoring infrastructure provides real-time visibility into system performance and enables rapid detection of any operational anomalies. The monitoring architecture consists of multiple integrated layers. At the lowest level, each Fronius inverter includes embedded data logging and communications capabilities, continuously recording AC power output, DC input voltage and current, operating temperature, and fault codes. This data transmits via Ethernet to a central data logger located in the main AC combiner enclosure.

Environmental monitoring includes a weather station mounted on a dedicated pole within the array, measuring global horizontal irradiance, module plane-of-array irradiance, ambient temperature, module back-of-module temperature, wind speed, and wind direction. This meteorological data enables performance ratio calculations and helps distinguish between production losses due to weather versus equipment issues.

A revenue-grade production meter, certified to ANSI C12.20 accuracy standards, measures all energy exported to PG&E's system. This meter, installed on the load side of the pad-mounted transformer, provides the authoritative record of production for both billing purposes (PG&E compensates Pacific Coast Renewables at wholesale rates) and for subscriber allocation through our community solar program software platform.

All monitoring data aggregates at a site controller (a ruggedized industrial PC) that hosts Solar-Log monitoring software and provides data connectivity via cellular modem to Pacific Coast Renewables' network operations center. The system generates automated alerts for fault conditions, communicating via email and SMS to on-call technicians. Historical data retention supports performance warranty verification, operations and maintenance planning, and subscriber reporting requirements.

Construction Sequencing and Coordination

Electrical construction activities integrate within the overall project schedule to optimize workflow and minimize duration. Following completion of site civil work and racking installation, electrical work proceeds in the following general sequence:

Underground conduit installation begins with layout and trenching for the AC collection system, transformer pad area, and utility interconnection route. Trenching operations will be carefully coordinated with the County's encroachment permit requirements and will include proper notification to USA North (811) for utility locates. The directional bore under the park access road requires specialized equipment and will be performed by Bay Area Boring, a subcontractor with extensive experience in HDD operations for sensitive locations.

DC combiner box installation and string wiring proceeds concurrently with module installation, allowing electricians to begin wiring strings as soon as racking sections are complete. This parallel workflow accelerates

the overall schedule while maintaining quality control—each string undergoes insulation resistance (megger) testing and polarity verification before connection to combiner boxes. String fuse sizing and installation follows manufacturer specifications, with all combiner boxes undergoing thorough inspection prior to inverter energization.

Inverter mounting, termination, and testing represents a critical milestone. Each inverter undergoes factory acceptance testing prior to shipment, but site acceptance testing verifies proper installation and programming. Tests include verification of DC input voltage and polarity, AC output voltage and phase rotation, ground continuity, proper operation of disconnects, and confirmation of monitoring communications. Only after successful completion of these tests do we proceed with energization.

PG&E interconnection represents the final construction phase and follows their standard commissioning process. This includes witness testing of the protection relay, verification of anti-islanding functions, confirmation of voltage and frequency trip settings, and validation of communication systems if applicable. Upon satisfactory completion of all tests and receipt of PG&E's Permission to Operate (PTO) letter, the system transitions to commercial operation.

Decommissioning and Site Restoration

Though the Harvey West array is designed for at least 30 years of productive operation, responsible project planning includes end-of-life considerations. Our decommissioning plan, required by County of Santa Cruz policy and secured by a $95,000 surety bond, ensures resources are available for complete site restoration when the project reaches end-of-life or if operations cease for any reason.

Decommissioning activities will include removal of all above-ground equipment (modules, racking, inverters, transformers), extraction of driven piles to a depth of 3 feet below grade, removal of all underground wiring and conduit to the extent practicable, and restoration of disturbed areas with appropriate topsoil and native vegetation. All materials will be recycled or disposed of in accordance with applicable regulations, with particular attention to solar module recycling through programs operated by the Solar Energy Industries Association (SEIA).

Contact Information Summary

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Contractor License: C-10 #E-876543

Electrical Engineer:
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Professional Electrical Engineer
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Property Owner:
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This document is part of a complete permit application package submitted to Santa Cruz County. Additional supporting materials including stamped engineering drawings, equipment cut sheets, geotechnical reports, and utility correspondence are provided under separate cover.